

Data Science @ scale 3/3

MODEL PERFORMANCE MONITORING

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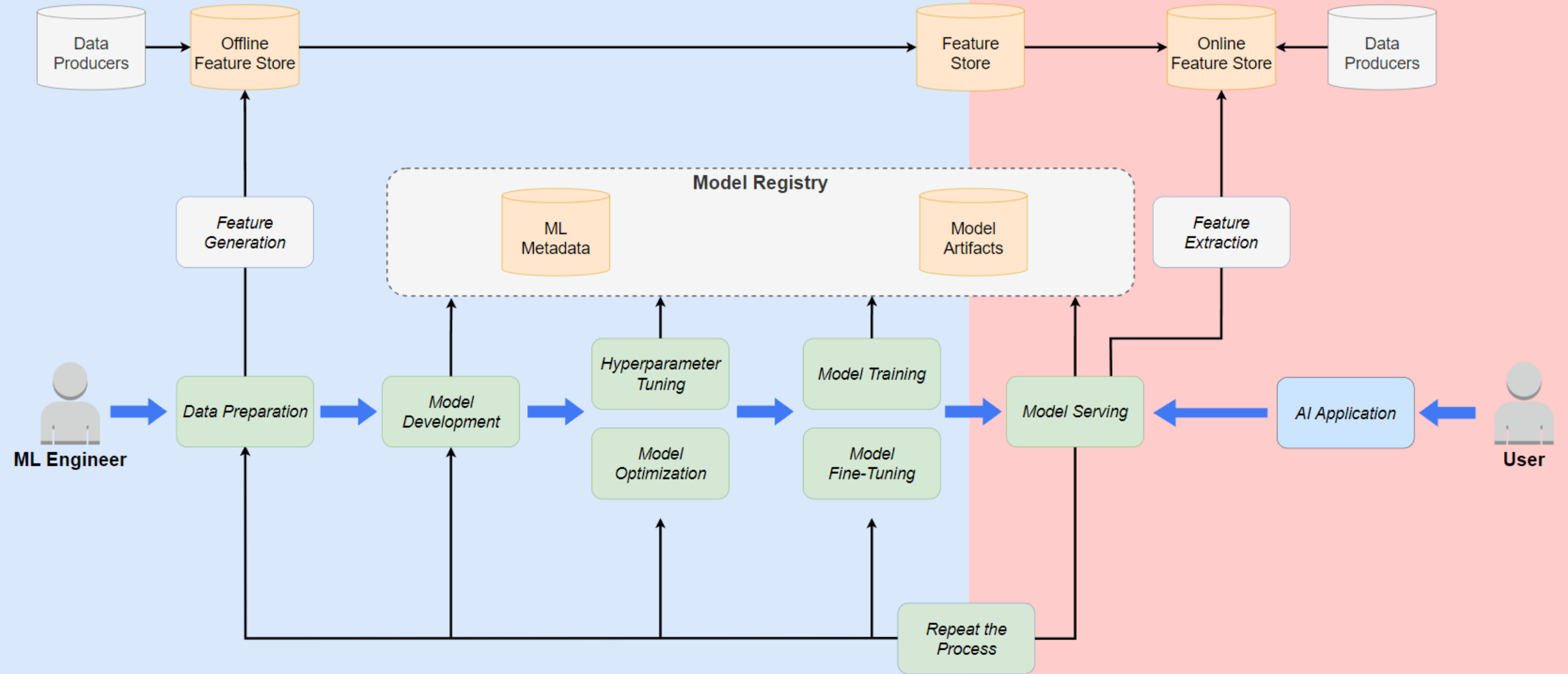
Motivation

- ▶ Part 1 was about inner loop: **exploration and development**
- ▶ Part 2 entered the **outer loop** with delivering properly
- ▶ Part 3 will be **model performance within outer loop**

- ▶ Key message: **outer loop also for data scientists : tests, explainability**

Development Phase

Production Phase



Governance & Responsible AI

- ▶ Model documentation (Model Cards, Data Sheets, Performances In and Out of sample)
- ▶ Ensure no bias (run on sub cohorts)
- ▶ Regulatory constraints (GDPR, auditability, anonymisation)
- ▶ We have just documented the model, this is not bullet proof...

A day in production

- ▶ User: the model prediction looks bad
- ▶ You: What was the prediction input ?
- ▶ User: Have you changed something ?, worked fine 1w ago
- ▶ You: Ok got it, let me try to explain the prediction
- ▶ You: Ok looks like a real issue, I'll check our feature store and see if a problem isn't even more serious, have you had more like this lately ?
- ▶ Fictitious but realistic. Time to log, Time to explanation. Time to local diagnostic, Time to real impact, Time to mitigation

Possible sources of error

- ▶ Datasources not updated (rolling window computations, one work and not the other...) due to some slow processing somewhere
- ▶ Data missing (api not reachable for instance) and missingness processing reveals to be bad
- ▶ Outlier badly handled, New modality in a categorical feature
- ▶ This can be a data problem but you have to debug: on a subset or more commonly through a **local explanation**

Local Explanation

- ▶ There are various techniques
- ▶ Each has its flaws and do not aim to reconstruct model reasoning
- ▶ Idea : attribute the outcome to some features more than others

Local Surrogates Models

- ▶ Idea: train a simple "surrogate" model for the neighborhood instance
- ▶ LIME: fit linear model from synthetic points in the neighborhood
- ▶ Local Rule-Based surrogates: extract rules through decision tree or other techniques to include non linearities (MARS: splines, MASALA: adaptative region)
- ▶ Pro: Model Agnostic
- ▶ Con: sensitive to sampling -> random. For rules sweet spot to find between too few (accuracy loss) or too many (interpretability loss)

SHAP

- ▶ Shapley values: Game theory, average marginal contributions across all coalitions of players
- ▶ SHAP: SHapley Additive exPlanations
- ▶ Some Model-specific implementations: TreeSHAP, KernelSHAP

- ▶ Pro: Solution to the 4 "fairness" axioms (Accuracy, Missingness, Symmetry, Monotonicity)
- ▶ Con: Computationally expensive for large models (many partitions to compute)

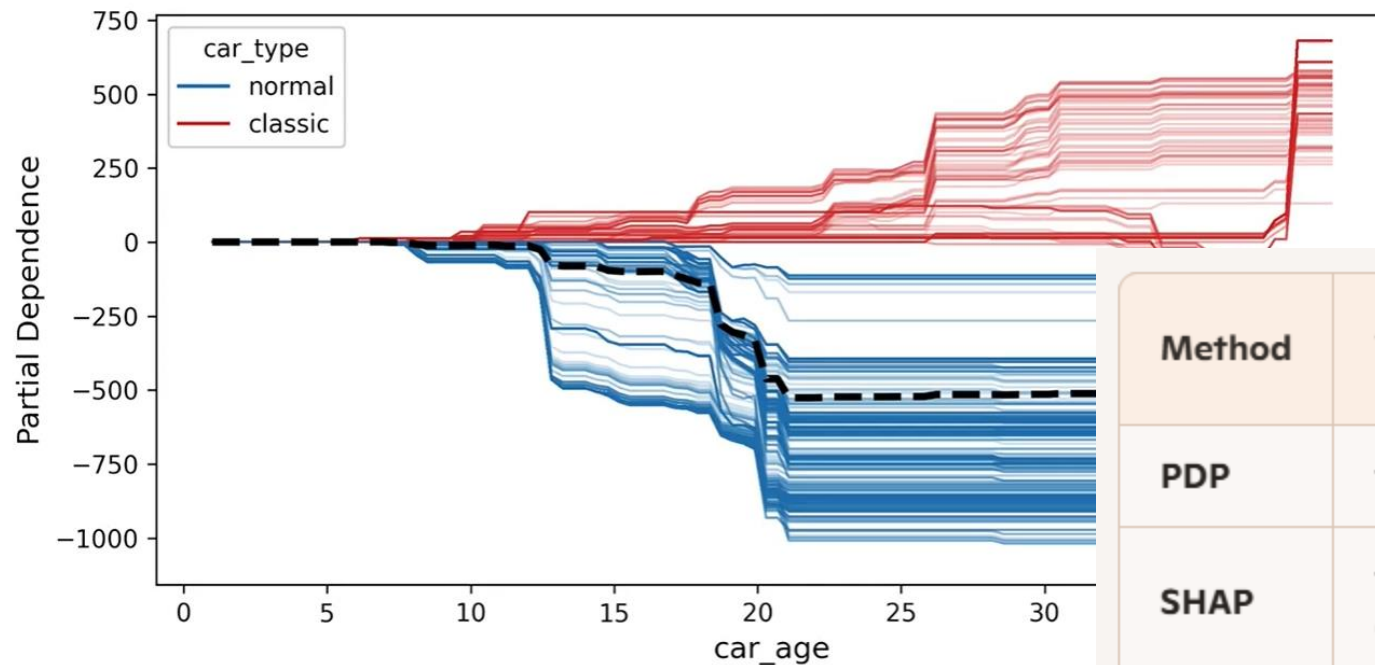
Gradient-Based Attributions

- ▶ How does the prediction change when I *move from a baseline input to the actual input*? Moving is progressively augment features to local point
 - ▶ Baseline is an input without information (white noise image or mean of the dataset)
 - ▶ Needs to have a gradient: good for Neural Networks (not trees, rules)
 - ▶ Integrated Gradients (IG): aggregate sensitivity along a straight path from baseline to your image
 - ▶ Saliency maps for images (pixel coloured depending)
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- ▶ Pro: Tailored for Model
 - ▶ Con: Just a sensitivity map

xAI: Explainable AI, not just local model

- ▶ Intrinsically Explainable Models: Decision Tree, (Generalized) Linear Models
- ▶ "Black Box" models: random forest, NN -> model agnostic
 - Global: Permutation Feature Importance, Partial Dependence Plot
 - Local (this prediction) -> what we discussed just before
- ▶ Causal models: remove proxy variable, find true cause
- ▶ Counterfactual explanations: what would change the prediction
- ▶ Adversarial examples: find inferences that defeat the model
- ▶ Non-agnostic methods

Partial Dependency Plot: Global Agnostic



Choose one feature and let all other fixed. Vary the feature and see how prediction changes.

Method	What it varies	What it averages	Type of effect
PDP	feature value	predictions	global
SHAP	feature presence (coalitions)	marginal contributions	local + global
IG	feature magnitude along a path	gradients	local

Permutation Importance: Global Agnostic

- ▶ For N sample, choose a feature
 - ▶ Permute the values of the feature among the N samples
 - ▶ See how it changes
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- ▶ Pro: works for any model
 - ▶ Con: The pb is that we can evaluate some impossible combinations, or very rare, not found in training set. This is true also with PDP

A last one: ALE

- ▶ Accumulated Local Effects
- ▶ Instead of averaging predictions over all data (like PDP), ALE computes **local differences** in small intervals of the feature.
- ▶ ALE only computes effects **within regions of the feature space where real data exists**, avoiding impossible combinations entirely.

From xAI to drifts

- ▶ xAI has limits: sovereign AI
- ▶ xAI is great but we would like to be able to be more proactive and anticipate problems before damage is done
- ▶ before the user gives feedback (and there may be no user at all but an agent)
- ▶ Because there can be some drift
- ▶ How to detect it ?

Rewind: Possible sources of error

- ▶ Datasources not updated (rolling window computations, one work and not the other...) due to some slow processing somewhere
- ▶ Data missing (api not reachable for instance) and missingness processing reveals to be bad
- ▶ Outlier badly handled, New modality in a categorical feature
- ▶ But irrespective of data quality, things may have really changed: this is called **drift**

Types of drift

- ▶ **Data drift** — raw input distribution changes: $P(X)$
 - ▶ **Feature drift** — engineered features degrade: X'
 - ▶ **Label drift** — target distribution changes: $P(Y)$
 - ▶ **Prediction drift** — model outputs shift: $P(Y | X, \text{model})$
 - ▶ **Concept drift** — relationship between X and Y changes $P(Y | X)$
 - ▶ **Model performance drift**: metric like accuracy degrades
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- ▶ The most important are Data, Concept.
 - ▶ Model performance is likely due to at least one of the 2

Operational Constraints

- ▶ Data regime: batch or online (streaming) ?
- ▶ Will ground truth be available and when ?
- ▶ Are we more interested in detecting large drifts or small progressive ?
- ▶ Do we want to be very sensitive but with many false positives ?
- ▶ What is acceptable ?
- ▶ Can we redeploy easily to retrain on newer data ?

Data Drift: batch stats on features

- ▶ Wait to have sufficient prod features for confident distribution comparison
- ▶ Choose a method to ascertain a statistical significant change
- ▶ Choose a method to compute drift
- ▶ Choose thresholds
- ▶ Determine if drift persists on a few periods

Adjusted p-values

- ▶ If you have many samples and feature changes you can be confident there is statistical significance for change
- ▶ If you consider many features in the dataset, you must adjust p-value for the nb of features tested (otherwise too many False Positives):
- ▶ **Family-Wise Error Rate (FWER)**: probability of *any* false positive: **Bonferroni** (to keep only big drifts),
- ▶ **False Discovery Rate (FDR)**: expected *proportion* of false positives among the flagged ones: **Benjamini-Hochberg** if you want more recall but with more false positive

Effect Size

- ▶ Significant OK but is this effect really important?: we call that effect size
- ▶ We take as effect size several statistical tests (KS test, PSI, divergences: KL, JS) and apply thresholds (from industry or calibrated on a reference period)

Metric	What it measures	Thresholds that matter
PSI	Binned distribution shift	0.1 mild, 0.25 investigate, 0.5 retrain
KS	Max difference in CDF	No universal threshold; use p < 0.01 but combine with effect size
Chi-square	Categorical drift	p < 0.01 + minimum expected counts
MMD	High-dimensional drift	Threshold must be calibrated on historical data

Data D: model based detection

- ▶ Multi-variate distance: Mahalanobis, MMD <- Classical anomaly detection
- ▶ Train encoder: compare embedding
- ▶ Train autoencoder on reference, compute reconstruction error on prod

Calibration and recalibration

- ▶ Why calibration matters (confidence scores, risk models)
- ▶ Calibration methods:
 - ▶ Platt scaling
 - ▶ Isotonic regression
 - ▶ Temperature scaling
- ▶ Recalibration can be sufficient, not necessary to retrain every time

Concept+Data: model performance

- ▶ Idea: if you have delayed true labels try to monitor full model performance
- ▶ Monitor accuracy, error rate, try to build a model from data drift per feature to model performance impact
- ▶ Be careful: performance is not linear in drift, performance can degrade long after drift (threshold -> see calibration)
- ▶ More globally: some drifts may not alter performance at all, it depends on feature importance, 2 drifts can also cancel out.
- ▶ You can check also if Feature Importance has varied

Drift Libraries

- ▶ 2 open source libraries: deepchecks and evidently. They provide many other checks on data quality
- ▶ Evidently offers commercial products like dashboards, LLM ...
- ▶ <https://www.evidentlyai.com/ml-in-production/data-drift>
- ▶ <https://www.evidentlyai.com/blog/embedding-drift-detection>

Continuous (online) detection

- ▶ ADWIN: Adaptive Window, looks for drift within the window
- ▶ DDM: Monitors Error rate
- ▶ Early DDM: DDM taking into account gradual change
- ▶ Software: River

RAG

- ▶ **RAGAS** (Retrieval-Augmented Generation Assessment) with LLM as judge
RAGAS takes **(question, retrieved documents, generated answer)** and returns **scores between 0 and 1** that quantify:
 - Did we fetch the right information ? (Contextual Relevance)
 - Did the LLM stay grounded in that information? (Faithfulness)
 - Is the answer relevant and complete? (Answer Relevance)
- ▶ No label required

Include feedbacks

- ▶ Have automated user feedback to retrain actively
- ▶ Conduct online experiments on clients (marketing): A/B testing, multi-armed bandits are run by Digital Experience Platforms (DXP) like Optimizely, VWO to get feedback
- ▶ DoE: Design of experiments
- ▶ Interleaving rankers: instead of A/B evaluate all rankers at once
- ▶ [A/B Testing for ML Models - Complete Guide | AI Wiki](#)

Last slide: look at some solutions

- ▶ Dataiku
- ▶ Weights and Bias
- ▶ Cloud vendor solutions: SageMaker, Vertex, DataBricks, Azure
- ▶ **Each has its pros and cons but we find globally the same features to help on the full lifecycle and serve all roles around data**